B. Extended related work

Compositional generalization. Research on compositional generalization investigates how models can systematically combine concepts. On the objective side, approaches such as Compositional Feature Alignment (Wang, 2025) and Compositional Risk Minimization (Mahajan et al., 2025) study how model training objectives, and model architecture (Jarvis et al., 2024) affect compositional generalization. On the representational side, kernel analyses characterize when certain compositional structures in embeddings yield generalization theoretically (Lippl & Stachenfeld, 2025), and empirical work investigates the role of disentangled representations for compositional generalization (Montero et al., 2021; Dittadi et al., 2021; Liang et al., 2025). On the data side, recent work probes whether and how scaling and data coverage improve compositional behavior (Uselis et al., 2025; Schott et al., 2022; Kempf et al., 2025). Abbasi et al. (2024) investigate CLIP's ability to recognize unlikely attribute-object combinations, finding that CLIP models still fall short on such tasks.

Other works establish formal sufficient conditions for when particular model classes can achieve compositional generalization, e.g., generative models whose data are produced by a differentiable rendering process and whose training distribution provides compositional support over latent factors (Wiedemer et al., 2023), constrained decoders paired with explicit inversion (Brady et al., 2025), conditional diffusion models with local conditional scores (Bradley, 2025), discriminative models whose inputs are drawn from an additive energy distribution (Mahajan et al., 2025), or linearly factorized representations (Uselis et al., 2025). In contrast, we do not impose specific structure on the data-generating process or on the learned representations. Instead, we ask what properties are implied *if* a model transfers from a restricted subset of the data space to the full space under our desiderata. Within this setting, our results can be interpreted as providing *necessary* conditions for compositional generalization for models that satisfy these desiderata.

Geometry of learned representations. A large literature studies the shape of learned features. In VLMs, Trager et al. (2023) report compositional linear subspaces, while in LLMs the *Linear Representation Hypothesis* (LRH) is examined mechanistically and statistically (Jiang et al., 2024; Park et al., 2023). Extending LRH, Engels et al. (2025) show that features can be multi-dimensional rather than rank-1, and Roeder et al. (2020) analyze identifiability constraints. Sparse-autoencoder probes provide evidence for monosemantic or selectively remapped features in VLMs (Pach et al., 2025; Zaigrajew et al., 2025; Lim et al., 2025). Beyond nominal labels, ordinal/ordered concepts motivate the rankability of embeddings (Sonthalia et al., 2025). More broadly, capacity limits for embedding-based retrieval emphasize geometric bottlenecks (Weller et al., 2025a). (Elhage et al., 2022) investigated empirically how neural networks can represent more features than there are dimensions in two-layer auto-encoder models. They found a tendency to encode features near-orthogonally with respect to neurons. Abbasi et al. (2024) find evidence of disentanglement in CLIP models. In contrast to these works, which document linear or near-orthogonal structure empirically, we show that under practice-driven desiderata and standard training, linearity and orthogonality are *necessary*.

Concurrent to our study of compositional generalization constraints in embedding spaces, Fel et al. (2025) conduct a large-scale, empirical concept analysis of DINOv2 using sparse autoencoders, showing strong task-dependent structure in which different downstream tasks recruit different concept families. They further propose a "Minkowski Representation Hypothesis", in which token activations behave as a Minkowski sum of convex polytopes, emphasizing both the interpretability consequences and the non-uniqueness/identifiability limitations of such decompositions. While their focus is token-level geometry and interpretability (rather than CLS-centric compositional probing and necessary conditions), their observations offer complementary evidence that modern vision encoders exhibit rich internal geometric structure beyond a purely unstructured embedding space.

Data, objectives, and training effects on geometry. Data distribution strongly shapes zero-shot behavior; concept frequency during pretraining predicts multimodal performance (Udandarao et al., 2024). On the objective side, BCE vs. CE can induce different feature geometries (Li et al., 2025), and contrastive/InfoNCE objectives exhibit characteristic similarity patterns (Lee et al., 2025). Convergence perspectives argue that the *objective* drives canonical representational forms (Huh et al., 2024), and objective choice has been tied to representational similarity across datasets (Ciernik et al., 2025).

Binding, explicit structure injection, and concept identification. Work on *binding* asks whether models maintain factored world states (Feng et al., 2025), and CLIP has been observed to show uni-modal binding (Koishigarina et al., 2025). Surveys and empirical studies examine binding limits and emergent symbolic mechanisms (Campbell et al., 2025; Assouel et al., 2025). Other approaches inject structure directly, e.g., hyperbolic image-text embeddings and entailment learning (Pal et al., 2024; Desai et al., 2024), or pursue concept identification at the causal/foundation interface and object-centric pipelines (Rajendran et al., 2024; Mamaghan et al., 2024).

Relation to disentangled and object-centric representations. Work on disentangled representations largely focuses on specifying desiderata for internal codes (e.g., disentanglement, completeness, informativeness) and proposing metrics or training schemes to satisfy them, often with the informal motivation that such structure should help downstream generalization (Bengio et al., 2014; Eastwood & Williams, 2018; Higgins et al., 2018). Closely related, object-centric representation learning injects an inductive bias that scenes should be modeled as compositions of objects, which can be viewed as a structured form of factorization/binding that may support compositional transfer and robustness (Greff et al., 2020). Recent studies directly probe how these properties relate to out-of-distribution or compositional generalization, often with mixed or limited evidence (Watters et al., 2019; Dittadi et al., 2021; 2022; Montero et al., 2021; 2024; Kapl et al., 2026). We instead ask a complementary question: if a discriminative model *does* exhibit compositional generalization when learned from a subset of the data space, what must necessarily be true of its embeddings?

A large body of literature has studied the usefulness and implications of learning disentangled representations in an unsupervised way (Bengio et al., 2014; Lake et al., 2017). Most commonly, the goal is to learn a generative model, usually through a VAE (Kingma & Welling, 2014), that can compress the data in a disentangled manner, in a way that allows to reconstruct these representations. While shown to be impossible without additional assumptions (Locatello et al., 2019), under weak supervision learning is possible (Shu et al., 2020; Locatello et al., 2020). Measuring the degree of disentanglement in these models is in itself non-trivial and various metrics have been proposed, e.g. by measuring disentanglement by performing interventions on the representations (Higgins et al., 2017; Kim & Mnih, 2018).

The DCI framework (Eastwood & Williams, 2018) proposes desiderata of properties disentangled representations should satisfy, namely disentanglement, completeness, and informativeness, and proposes a metric to measure them. Some works also consider what constitutes a good disentanglement (Higgins et al., 2018) and propose a conceptual framing of meaning behind disentangled representations with respect to the data generative process in terms of group actions of transformations.

Abbasi et al. (2024) investigate the role of representation disentanglement in compositional generalization in CLIP models. Using metrics such as DCI, they find that CLIP models with more disentangled text and image representations exhibit higher compositional OOD accuracy on their attribute-object dataset (ImageNet-AO). This work is complementary to ours. Their study explores correlations between disentanglement and compositional generalization by probing CLIP embeddings with respect to the adjective and noun components present in the inputs. For instance, they estimate "attribute" and "object" subspaces by feeding isolated adjectives or nouns into the text encoder, or by generating isolated attributes/objects via a text-to-image model and embedding them with CLIP. However, this approach assumes that CLIP's embedding space is additively decomposed with respect to individual words, an assumption that is not guaranteed to hold. Indeed, Yamada et al. (2024) show that word embeddings in language models are often highly entangled with associated concepts. In contrast, our necessary condition does not rely on word-level decomposition. We posit that models achieving perfect downstream compositional performance must possess linearly factorized representations that separate per-concept components, independent of how an encoder processes individual words. In short, our work provides principled motivation for analyses of representational decomposition, whereas Abbasi et al. (2024) offer an empirical correlation study based on CLIP's emergent disentanglement.

Lippl & Stachenfeld (2025) investigate when a particular form of compositionally structured representations, specifically representations whose similarity depends only on how many underlying components two inputs share, supports downstream compositional generalization. Using kernel theory, they characterize exactly which tasks linear readouts on top of such representations can solve, showing that these models are fundamentally restricted to conjunction-wise additive functions. In contrast, we focus on a specific subclass of compositional tasks: identifying factors of inputs that never co-occur during training. While Lippl & Stachenfeld (2025) characterize what kinds of generalization are possible under a compositional representational structure, we ask the complementary question: given perfect downstream performance on such a task, what representational structure must the model necessarily possess under the desiderata we specify?

C. Necessary conditions (proof of Proposition 1)

In this section, we prove Proposition 1 and state the auxiliary lemmas used in the argument.

Our default training-support regime is the one used in the main result: valid supports satisfy $|T| = 2^{k-1} + 1$. In the proof below, we work with cross-datasets $\mathcal{D}^{\boldsymbol{c}}$ (Definition 6), i.e., datasets centered at $\boldsymbol{c}$ that vary one concept at a time, because this makes the core geometric argument transparent. We write $\mathcal{D}$ for the full dataset of all n^k combinations and $\mathcal{D}^{\boldsymbol{c}}$ for a cross-dataset centered at $\boldsymbol{c}$.

Although most proofs in this section are written in the binary setting for clarity, we state intermediate claims in a form that extends directly to general n where possible. Concretely, in the binary setting $\mathcal{C}_i = \{0, 1\}$. Any learned quantity carries a superscript indicating the training set, e.g., $\boldsymbol{w}_i^{(\mathcal{D})}$ or $\boldsymbol{w}_i^{(\mathcal{D}^{\boldsymbol{c}})}$. For concept i and training set S, we use the logistic score form

$$h_i^{(S)}(\boldsymbol{z}) := (\boldsymbol{w}_i^{(S)})^\intercal \boldsymbol{z} + b_i^{(S)}, \qquad p_i^{(S)}(\boldsymbol{z}) = \sigma(h_i^{(S)}(\boldsymbol{z})),$$

and we drop the superscript when the dataset is clear from context.

Definition 5 (Intervention on a concept value)**.** For any concept index $i \in [k]$, target value $j \in [n]$, and concept vector $\boldsymbol{c} \in [n]^k$, we define the intervened index and representation

$$\boldsymbol{c}(i \to j) := (c_1, \ldots, c_{i-1}, j, c_{i+1}, \ldots, c_k), \quad \boldsymbol{z}_{\boldsymbol{c}(i\to j)} := \boldsymbol{z}_{\boldsymbol{c}} \text{ with concept } i \text{ set to } j.$$

In case of multiple interventions, they compose componentwise, e.g. for distinct $i, m \in [k]$, $\boldsymbol{c}(i \to j, m \to l) = (c_1, \ldots, c_{i-1}, j, c_{i+1}, \ldots, c_{m-1}, l, c_{m+1}, \ldots, c_k)$.

Definition 6 (Cross dataset at $\boldsymbol{c}$)**.** Given a concept space $\mathcal{C} = \mathcal{C}_1 \times \cdots \times \mathcal{C}_k$, we say that a dataset $\mathcal{D}^{\boldsymbol{c}}$ is a cross-dataset at $\boldsymbol{c} \in [n]^k$ if:

1. It contains only samples that vary one concept at a time around the center $\boldsymbol{c}$:

$$\mathcal{D}^{\boldsymbol{c}} = \big\{(c_1', c_2, \ldots, c_k) : c_1' \in [n]\big\} \cup \cdots \cup \big\{(c_1, c_2, \ldots, c_{k-1}, c_k') : c_k' \in [n]\big\} = \{\boldsymbol{c}\} \cup \bigcup_{i=1}^{k} \bigcup_{a \in [n]\setminus\{c_i\}} \{\boldsymbol{c}(i \to a)\}.$$

2. Its size is $1 + k(n-1)$,

Definition 7 (Dataset marginal counts)**.** For any dataset $S \subseteq \{(\boldsymbol{z}_{\boldsymbol{c}}) : \boldsymbol{c} \in [n]^k\}$ (e.g., $S = \mathcal{D}$ or $S = \mathcal{D}^{\boldsymbol{c}}$), concept $i \in [k]$, and value $j \in [n]$, the marginal count of value j in S is

$$N_{i,j}(S) := \big|\{\boldsymbol{c} \in [n]^k : (\boldsymbol{z}_{\boldsymbol{c}}) \in S, c_i = j\}\big|.$$

When S is clear, we abbreviate $N_{i,j} := N_{i,j}(S)$.

Remark 1 (Marginal counts: full vs cross-datasets)**.** For the full dataset $\mathcal{D}$, the marginal counts are balanced:

$$N_{i,j}(\mathcal{D}) = n^{k-1} \quad \text{for all } i \in [k],\ j \in [n].$$

For a cross-dataset $\mathcal{D}^{\boldsymbol{c}}$ as in Definition 6, the marginal counts satisfy

$$N_{i,j}(\mathcal{D}^{\boldsymbol{c}}) = \begin{cases} 1 + (k-1)(n-1), & j = c_i, \\ 1, & j \neq c_i. \end{cases}$$

Proof. In $\mathcal{D}$, fixing $c_i' = j$ leaves n^{k-1} free coordinates. In $\mathcal{D}^{\boldsymbol{c}}$: varying concept i contributes one point for each $j \neq c_i$; the center contributes one more with $c_i' = c_i$; varying any other concept $r \neq i$ adds $(n-1)$ points with $c_i' = c_i$, across $(k-1)$ such concepts, totaling $(k-1)(n-1)$. □

Definition 8 (Binary complement notation)**.** In the binary case ($\mathcal{C}_i = \{0, 1\}$), we write $\bar{c}_i := 1 - c_i$ for the complement value of concept i. As shorthand for an intervention to the complement, we write $\boldsymbol{c}(\bar{c}_i) := \boldsymbol{c}(i \leftarrow \bar{c}_i)$.

To make use of Stability in the binary case, we use the identifiability of the sigmoid.

Lemma 1 (Binary equal probabilities imply equal affine parameters)**.** For a concept index i and two training supports S, S', let

$$h_i^{(S)}(\boldsymbol{z}) := (\boldsymbol{w}_i^{(S)})^\intercal \boldsymbol{z} + b_i^{(S)}, \quad h_i^{(S')}(\boldsymbol{z}) := (\boldsymbol{w}_i^{(S')})^\intercal \boldsymbol{z} + b_i^{(S')}.$$

Assume that for every $\boldsymbol{z} \in \mathbb{R}^d$,

$$\sigma\Big(h_i^{(S)}(\boldsymbol{z})\Big) = \sigma\Big(h_i^{(S')}(\boldsymbol{z})\Big),$$

where $\sigma(t) = 1/(1 + e^{-t})$. Then

$$\boldsymbol{w}_i^{(S)} = \boldsymbol{w}_i^{(S')} \qquad \text{and} \qquad b_i^{(S)} = b_i^{(S')}.$$